

BALAJI RADHAKRISHNAN PADMANABHAN

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SUMMARY

Software Development Engineer in Test (SDET) with 3 years of experience in test automation, API testing, and quality assurance. Proficient in Java, Selenium WebDriver, Cucumber BDD, REST Assured, and CI/CD pipelines. Currently pursuing Master of Science in Computer Software Engineering at Arizona State University. Seeking Job opportunities in SDET and QA Automation roles.

EDUCATION

Master of Science in Computer Software Engineering <i>Arizona State University, IRA Fulton School of Engineering GPA: 3.33</i>	Aug 2025 – May 2027 Tempe, AZ
Bachelor of Technology in Computer Science and Engineering (AI & ML) <i>SRM Institute of Science and Technology</i>	June 2018 – May 2022 Chennai, India

TECHNICAL SKILLS

- **Programming & Scripting:** Java, JavaScript, Python, SQL, Gherkin, HTML/CSS
- **Test Automation & Frameworks:** Selenium WebDriver, Cucumber BDD, TestNG, JUnit 5, REST Assured, Postman, Appium, Page Object Model (POM), Selenium Grid, JMeter
- **CI/CD & Cloud:** Jenkins, Maven, Git, GitHub, AWS (EC2, S3, DynamoDB, Lambda, CodeBuild, CodePipeline, CloudWatch)
- **Databases & Methodologies:** Oracle, PostgreSQL, MySQL, DynamoDB; Agile (Scrum/Kanban), BDD/TDD, Test Plans & Strategies

PROFESSIONAL EXPERIENCE

Tata Consultancy Services <i>Software Development Engineer in Test (SDET)</i>	Sept 2022 – July 2025
• Designed, developed, and executed test plans, test cases, and automated scripts using Java, Selenium WebDriver, Cucumber BDD, and TestNG, ensuring comprehensive software quality across web applications. • Performed functional, regression, integration, and performance testing using JMeter, identifying 150+ critical defects and ensuring timely resolution through clear documentation and collaboration with developers. • Developed and maintained API test automation framework using REST Assured and Postman, validating RESTful endpoints, data accuracy, and backend integrations with Oracle and PostgreSQL databases. • Implemented Page Object Model (POM) design pattern and BDD practices to create maintainable, reusable test scripts, improving regression efficiency by 25% and reducing manual testing effort. • Configured and managed CI/CD pipelines using Jenkins and AWS CodeBuild/CodePipeline, enabling automated test execution on every build and deployment to EC2 and S3 environments. • Conducted cross-browser and cross-platform testing using Selenium Grid, ensuring consistent functionality across Chrome, Firefox, Safari, and Edge browsers. • Collaborated with developers, product managers, and stakeholders in Agile (Scrum) ceremonies to understand requirements, review specifications, and identify potential areas of concern before testing. • Utilized AWS services including DynamoDB, CloudWatch, Lambda, and Secrets Manager to support test data management, monitoring, and secure configuration handling.	

ACADEMIC PROJECTS

E-Commerce Test Automation Suite <i>Java, Selenium, Cucumber BDD, TestNG</i>	Fall 2025
• Developed comprehensive automated test suite for Amazon.com e-commerce platform using Java, Selenium WebDriver, Cucumber BDD, and TestNG framework. • Implemented Page Object Model (POM) design pattern to create maintainable, reusable test scripts covering search functionality, cart operations, and checkout flows. • Integrated REST Assured for API testing and configured Jenkins CI/CD pipeline for automated test execution on every commit.	
Healthcare Navigator <i>React, FastAPI, Python, GraphDB, MongoDB</i>	Fall 2025
• Built knowledge graph integrating CMS physician data, hospital ratings, and symptom mappings using OWL ontology and SPARQL queries. • Developed FastAPI backend with geospatial ranking (Haversine formula) and MongoDB caching for optimized query performance.	
Requirements Engineering Tool <i>React, FastAPI, Python, JavaScript, MySQL</i>	Fall 2025
• Developed full-stack web application using React frontend and FastAPI backend with MySQL database for Agile requirements lifecycle management. • Implemented features for user story authoring, backlog grooming, sprint readiness validation, and Jira export integration using Python and JavaScript. • Built role-based dashboards with React, focusing on accessible design patterns and clear content hierarchy.	